2. What are interpolation and extrapolation in numerical methods and why do we use them? What is the difference between them? Explain briefly. (5 pts)

Q2. Find Lagrange interpolating polynomials that approximate $f(x) = x^3$.

a. The linear interpolation polynomial $P_1(x)$ using the nodes $x_0 = -1$ and $x_1 = 0$. (4 pts)

b. The quadratic interpolation polynomial $P_2(x)$ using the nodes $x_0 = -1$, $x_1 = 0$, and $x_2 = 1$. (6 pts)

c. The cubic interpolation polynomial $P_3(x)$ using the nodes $x_0 = -1$, $x_1 = 0$, $x_2 = 1$, and $x_3 = 2$. (8 pts)

Answer:

3. The set of the following five data points is given:

x	1	2	4	5	7
y	54.	7.	-3	-38	12

a. Determine the fourth-order interpolation polynomial in Newton's form that passes through these points. Calculate the coefficients by using a divided difference table. (15 pts)

b. Use the polynomial obtained in part (a) to determine the interpolated value for **x=3**. (5 pts)

4. The following data shows estimates of the population of a city in selected years between 1970 and 2020:

Year	1970	1980	1990	2000	2010	2020
Population (millions)	1.1	1.4	1.9	2.1	2.8	4

a. Calculate the rate of growth of the population in millions per year for 2020 using the backward difference formula. (10 pts)

b. Using the slope in 2020 from part (a), apply the central difference formula to extrapolate and predict the population in the year **2030**. (10 pts)

5. Let $f(x) = xe^x$.

a. Calculate approximations to $f'(2.3)$ using the backward, forward, and central difference formulas with $h=0.1$, and $h=0.01$. (15 pts)

b. Compare the results with the exact (analytical) derivative (using absolute errors). Determine the best approximation. (5 pts)

④ The function $f(x)$ is given in the following tabulated form. Compute $\int_{?}^{1.8} f(x)dx$ with $h = 0.3$.

a) Use the composite left endpoint method.

b) Use the composite midpoint method

c) Use the composite trapezoidal method.

x	0	0.3	0.6	0.9	1.2	1.5	1.8
f(x)	0.5	0.6	0.8	1.3	2	3.2	4.8

3. The set of the following five data points is given:

x	1	2	4	5	7
y	52	5	-5	-40	10

a. Determine the fourth-order polynomial in Newton's form that passes through the points. Calculate the coefficients by using a divided difference table. (15p)

b. Use the polynomial obtained in part (a) to determine the interpolated value for **x=3**. (5p)